ARCH 305
Architecture and Town Planning

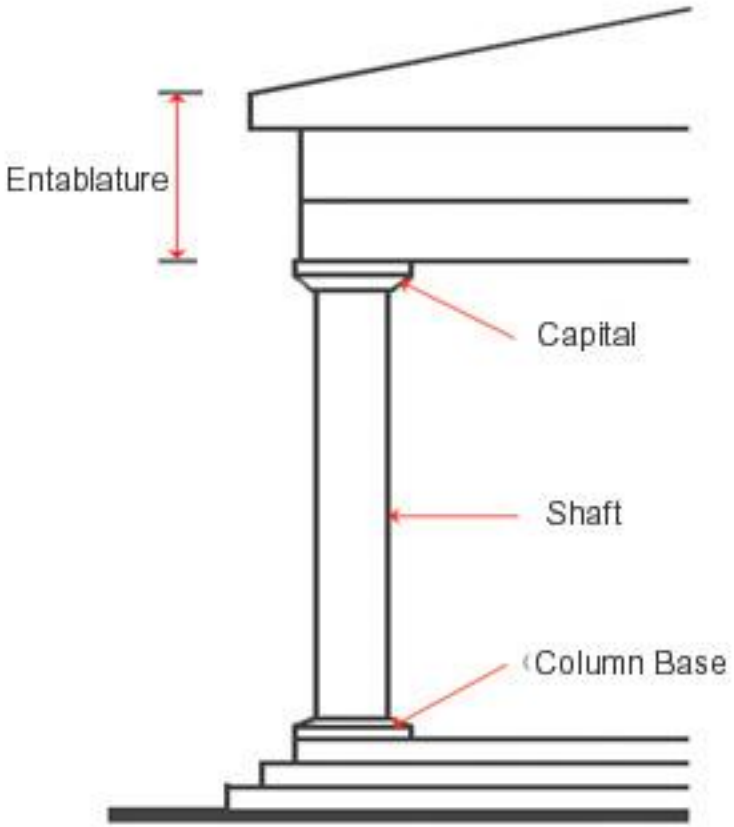
Ancient Architecture



Greek Architecture

The Orders
Introduction

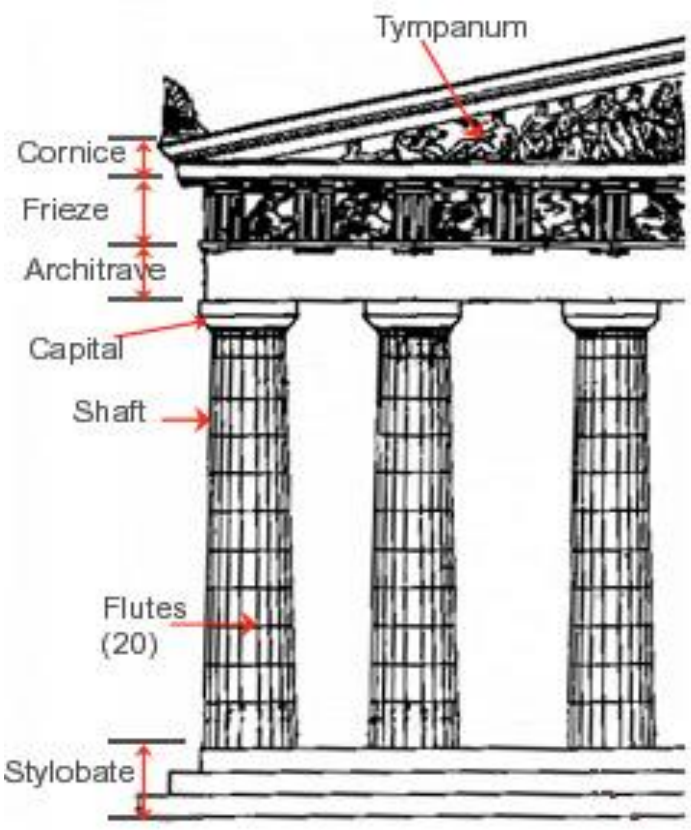
- Refer to the entire set of form that makes up the principal elevation of a temple.
- Composed of a base, an upright column or support with its capital, and the horizontal entablature.
- All the parts of an order are proportionally derived from the size of the base of the column.
- It determines all aspects of the elevation of a building including its shape and the arrangement and proportion of its parts



Compponents of the Greek Order

The Orders
Introduction

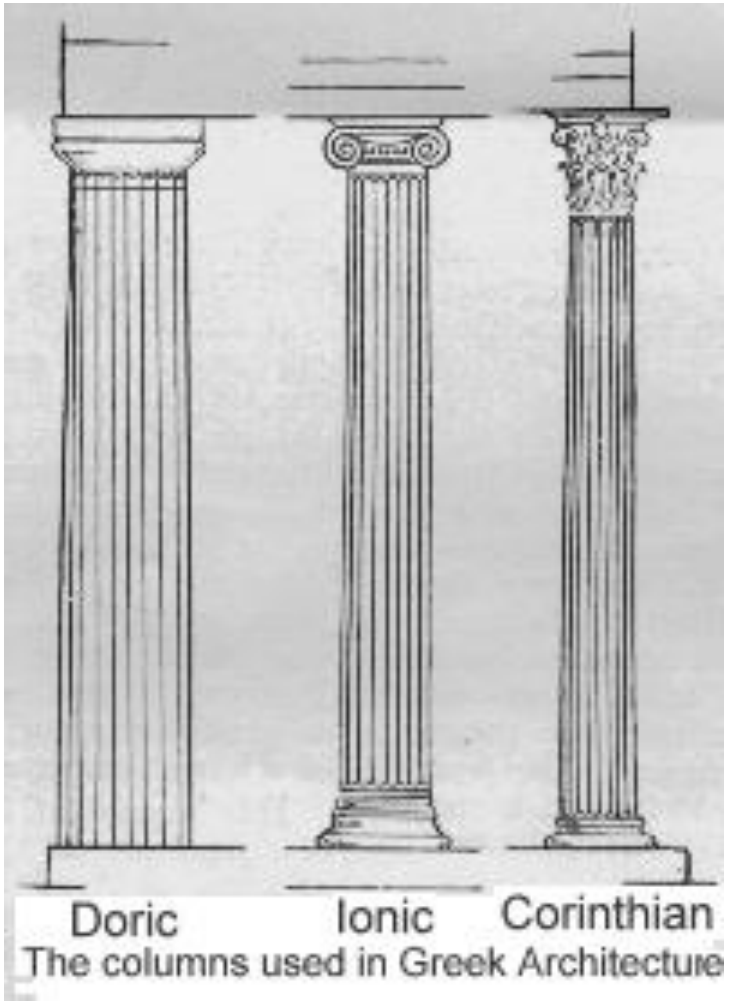
- The entablature is divided into three sections;
  - the cornices,
  - the frieze and
  - the architrave
- According the rules of classical architecture, the entablature should always be divisible into these three zones



The Doric Order
Source: Adapted from Sir Barnister Fletcher (1996)

The Orders
Introduction

- Greeks are credited with originating the three orders of the classical language of architecture, Doric, Ionic and Corinthian.
- Columns were understood by the Greeks to be anthropomorphic or representative of the body of a human
- The base suggests the feet, the shaft the torso and the capital the head.

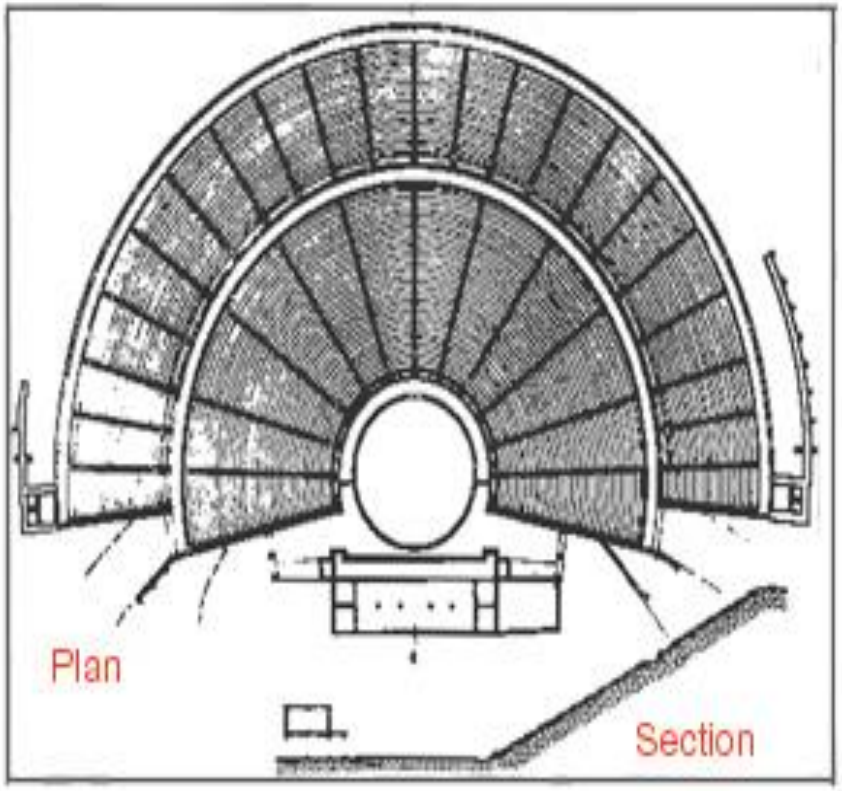


The columns used in Greek Architecture

Civic Architecture

Theaters – Theater Epidarus

- The Greeks invented the theater design that is still used in movies and auditoriums today
- Every important Greek city had a theater
- Their theater was built into a hilly landscape
- The theater had a bank of seats steps created from the landscape



Theater Epidaurus, Plan & Section

Greek City Planning and Design

Greek City Planning and Design

Planning and Design Principles

- The ancient Greek civilization had established principles for planning and designing cities
- City form were of two types
- Old cities such as Athens had irregular street plans reflecting their gradual organic development
- New cities, especially colonial cities established during the Hellenistic period, had a grid-iron street plan
- Certain things were common among cities

Greek City Planning and Design

Planning and Design Principles

- Towns had fixed boundaries and some were protected by fortifications
- Much of the town was devoted to public use
- The Greek City was usually divided into three parts; the acropolis, the agora and the town.
- Site planning and design was centered on the appreciation of buildings from the outside.
- The location of buildings was therefore such that it could command a good view to it.

Greek City Planning and Design

The Acropolis

- The Acropolis was the city of temples
- It is the location where all the major temples of a city are located
- It was built to glorify the gods
- Greeks considered high places to be important & sacred
- The Acropolis were usually located on the highest ground
- Other public buildings such as gymnasia, stadia, and theaters were generally regarded as part of religious rituals
- They are normally found attached on lower ground to the hills of the Acropolis

Greek City Planning and Design

The Agora

- The Agora was the most important gathering place in a Greek city
- It started as an open area where the council of the city met to take decisions
- With time buildings were constructed to define and enclose the space
- It also transformed into a place for combined social, commercial and political activities
- It emerged as the heart of Greek intellectual life and discourse.
- It was usually located on a flat ground for ease of communication
- It was placed to be easily accessible from all directions
- In many cities, it is also located close to the Acropolis

Greek City Planning and Design

The Town

- The town was where the people lived
- This was the domain of women, who did not have any public role
- Early Greek towns had an irregular street pattern, resulting from its organic growth
- Later Hellenistic towns such as Prienne had a formal rectilinear pattern
- The town was made up of only residential houses



Egypt Architecture

E G Y P T



EGYPTIAN PYRAMIDS: THE RAMP THEORY

One theory suggests that **RAMPS** were used to haul the **STONE BLOCKS** on **WOODEN SLEDS** up the side of the pyramids. The ramps were **LUBRICATED WITH WATER** to reduce friction when hauling the blocks. As few as 10 men were needed to drag a stone block up a ramp. may have been several ramps on each side of the pyramid at different levels, and a ramp may have been coiled around the pyramid as it grew in height. Once a stone block reached its desired level, wooden rockers may have been used to maneuver it into position.



Ramp on pyramid

E
G
Y
P
T



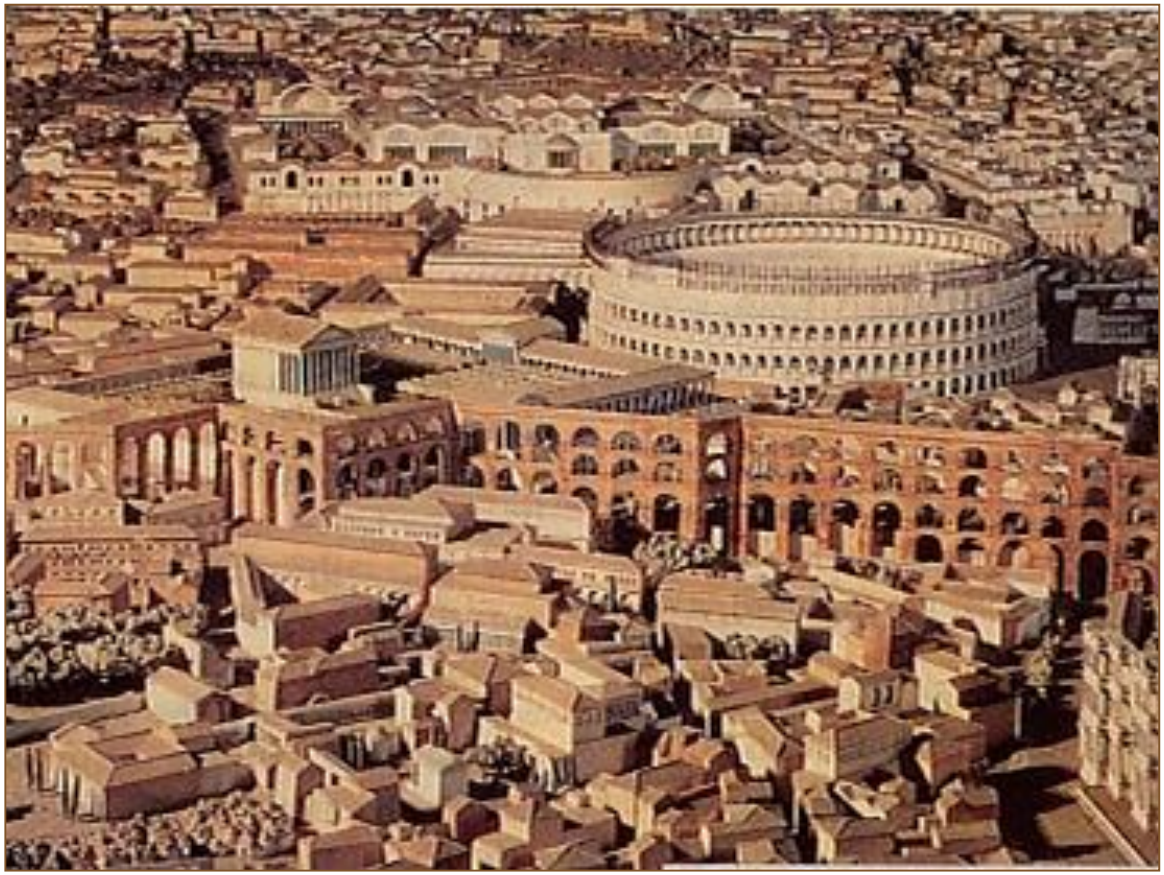
Stone block on sled



The pyramids were probably **NOT BUILT BY SLAVES** because slave labor was not widely used in Egypt at the time. **PEASANT FARMERS**, however, were required to spend a number of weeks working on construction projects. This provided the paid labor needed to build these gigantic structures. Since the fields were under water during the summer, wages earned in building the gigantic pyramids **SUPPLEMENTED THE FAMILY'S INCOME**.



Pyramids did not stand alone; they were part of a **FUNERARY COMPLEX**. The complex includes a **PROCESSIONAL CAUSEWAY** that links a **FUNERARY TEMPLE** to the pyramid.



Roman Architecture

R

O

M

E

ENGINEERING PRACTICES

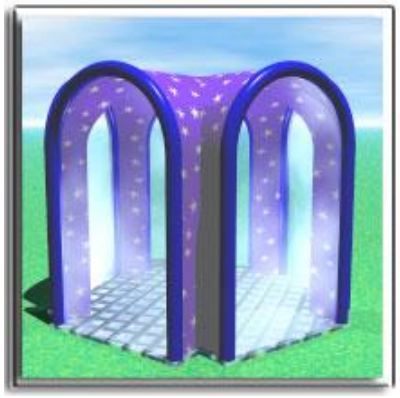
- The practical nature of the Romans and their skill and initiative in engineering were demonstrated in the many **DAMS, RESERVOIRS**, and **HARBORS** they built.

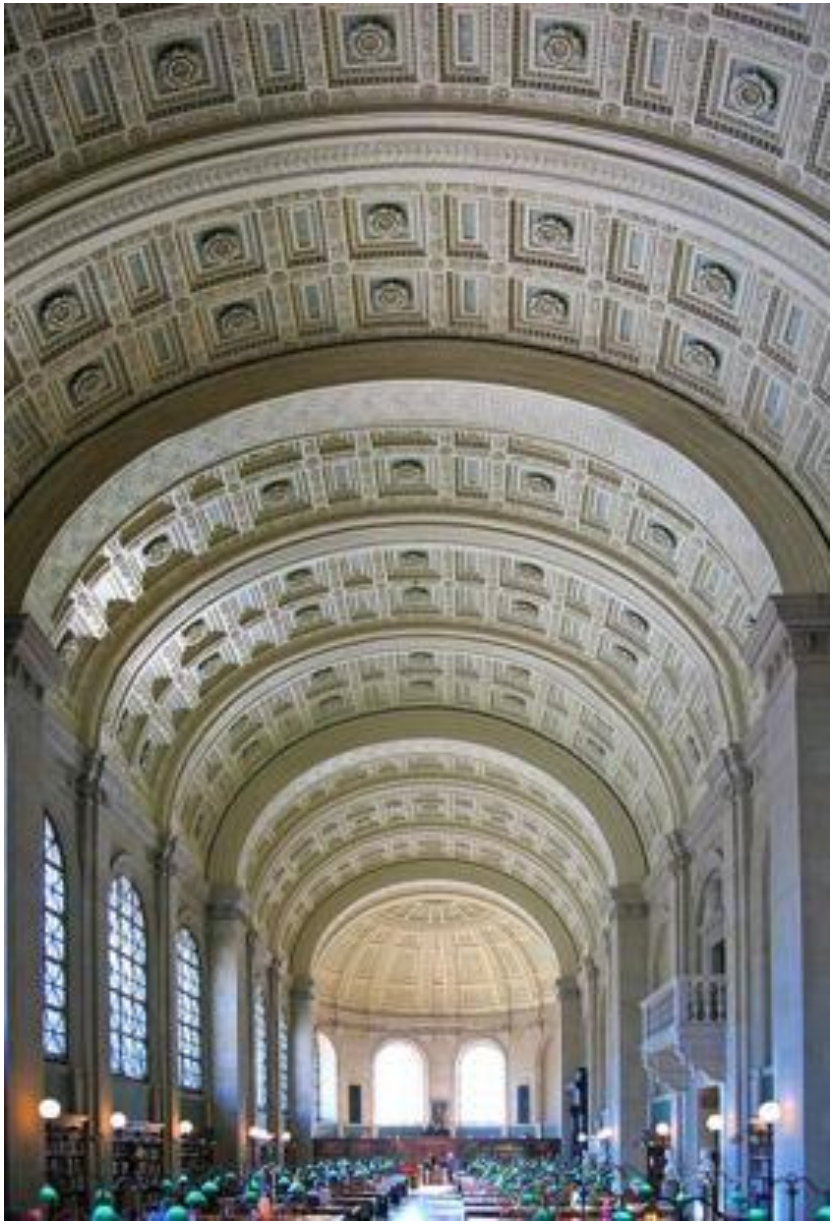
- In designing their bridges and aqueducts, the Romans placed a series of **STONE ARCHES** next to one another to provide mutual support.



ROME

- The **BARREL VAULT**, basically a series of adjoining arches forming a structure resembling a tunnel, was a new method of enclosing space. In the barrel vault the supports of the arches became heavy masonry walls to bear the weight of the vaulted roof.
- The Romans next developed the **CROSS VALUT** by intersection two barrel vaults at right angles.





ROME

- Another important advance in architecture was the Roman's success in constructing **CONCRETE DOMES** on a large scale. The weight of the dome was transferred directly to the walls and no other support was necessary. The largest of the dome structures was the **PANTHEON** (temple of all the gods).





- The standard type of Roman public building was the **BASCILICA**, a colonnaded structure that became a model for early Christian churches. Rows of columns divided the interior into a central nave and side aisles, with the roof over the nave raised to admit light, creating a **CLERESTORY** (an upper portion of a wall containing windows for supplying natural light to a building.



Modern Architecture



What are the differences between traditional architecture and modern architecture?





	shape
ancient architecture	Like things we find in nature. Nature does not have any straight lines.
modern architecture	Huge, like boxes with flat roofs, sharp corners and glass walls.

	material
ancient architecture	earth, stone, brick, wood and bamboo
modern architecture	steel, glass, concrete, alnico and plastic

	feeling
ancient architecture	Natural and beautiful
modern architecture	Hard and unfriendly